

教學實作：擴充 SG90 伺服馬達功能

步驟 1：修改硬體定義提示詞 (Devices Definition)

讓 AI 認識伺服馬達及其物理規則。

- SG90 servo: pin 5
- servo angle control
 - range: 0–180
 - 0 = fully closed
 - 180 = fully open

步驟 2：定義工具 JSON 規範 (Tools Definition)

在 `tools_definition` 中加入 `/servo` 指令的格式規範。

Servo motor control:

```
{
  "type": "tool_call",
  "method": "/servo",
  "params": {
    "pin": "<Device pin number. If the user does not specify a pin, ask first.>",
    "angle": "<Desired absolute angle from 0 to 180>"
  }
}
```

Success response:

```
{
  "status": "success",
  "method": "/servo",
  "workId": "<system-provided>"
}
```

Error response:

```
{
  "status": "error",
  "method": "/servo",
  "reason": "<error reason>",
  "workId": "<system-provided>"
}
```

步驟 3：底層 C++ 程式碼實作

在專案中新增 `tool_servo` 驅動函數。

```
#include <AmebaServo.h>
AmebaServo servos[26];

String tool_servo(AmebaServo &servo, int pin, int angle, String workId) {
    if (!servo.attached())
        servo.attach(pin);

    if (angle < 0 || angle > 180) {
        return
            "{\"status\":\"error\","
            "\"method\":\"/servo\","
            "\"reason\":\"invalid_servo_angle\","
            "\"workId\":\"" + workId + "\"}";
    }

    servo.write(angle);

    return
        "{\"status\":\"success\","
        "\"method\":\"/servo\","
        "\"workId\":\"" + workId + "\"}";
}

void setup() {
    // ... 原有初始化內容 ...
}
```

步驟 4：更新工具分發器邏輯 (useTools)

將 /servo 方法註冊進系統的分發流程。

```
void executeTool(String workId, String command, JsonObject params, bool reCheck = true) {

    if (command == "/digitalwrite" || command == "/analogwrite") {
        // ... 原有內容不變 ...
    }

    else if (command == "/servo") {
        int pin    = params["pin"].as<int>();
        int angle = params["angle"].as<int>();

        String response = tool_servo(servos[pin], pin, angle, workId);

        if (xSemaphoreTake(stateMutex, MUTEX_TIMEOUT_TICKS) == pdTRUE) {
            historicalMessages += buildGeminiMessage("user", command + timestamps);
            historicalMessages += buildGeminiMessage("model", response + timestamps);
            executeToolHistory += workId + " " + command + " [ " + String(pin) + " | " + String(angle) + " ]\n";
            xSemaphoreGive(stateMutex);
        }

        evaluateWorkflowContinuation(workId, reCheck);

    }
    else if (command == "/digitalread" || command == "/analogread") {
        // ... 原有內容不變 ...
    }
    // ... 其餘 tools 不變 ...
}
```